

# Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



## Participant Details

- a. Participant Name: Anshul Jain (Team BloodWyrn, Solo)
- b. Problem Statement: AI for Better Living and Smarter Communities.  
Create a data intelligence tool people would actually use, and show how acceleration helps them make a faster or better decision.

## Brief about the idea

**VayuSense** is an AI decision intelligence platform that turns millions of raw air quality readings into one honest answer: ***is it safe to go outside right now?***

Built for parents, schools, clinics, and city officials across Indian and APAC cities.

**5.9M+**

real OpenAQ sensor readings, processed  
on GPU

**37.5×**

faster than pandas: NVIDIA RAPIDS cuDF  
benchmark

**2 agents**

Google ADK pipeline on Gemini: Analyst →  
Advisor

It goes beyond a raw AQI number: a **Safety Score**, a **Human Impact** estimate (cigarette-equivalent exposure, life-expectancy impact), and a plain-language verdict anyone can act on.

Fully deployed and live at [vayusense-kggf.onrender.com](https://vayusense-kggf.onrender.com).

Your solution should be able to explain the following:

- How did you approach the common problem statement and translate it into a working solution using Google Cloud technologies (e.g., BigQuery, ADK, Vertex AI, Gemini, etc.) while ensuring the hackathon's core requirements are met? If you selected the NVIDIA problem statement, also explain how NVIDIA technologies were integrated into your solution.
- What real-world problem does your solution address, and what practical impact does it create for users, organizations, or communities??
- What is the core architecture or workflow behind your solution, and how does it transform data into actionable insights or intelligent decisions?

### Approach

Ingest **5.9M+ OpenAQ readings**; clean and trend-analyze on GPU with **NVIDIA cuDF**; reason with **Google ADK agents** (Analyst → Advisor) on **Gemini**; ship as a deployed FastAPI app.

### Real-world impact

Delhi NCR breathes hazardous air most days. VayuSense turns raw data into **same-day decisions** for schools, clinics, and city wards; not another chart nobody reads.

### Architecture

OpenAQ → NVIDIA cuDF (GPU) → ADK/Gemini agents → FastAPI dashboard. Raw readings become a **Safety Score**, a **Human Impact** estimate, and a clear recommendation.

## Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

### Proven acceleration, not a claim

A reproducible **37.5× cuDF vs pandas benchmark** over 5.9M+ real rows, shown live inside the app.

### Human stakes, not just AQI

Cigarette-equivalent exposure and life-expectancy impact, grounded in published epidemiology.

### Agents, not a prompt wrapper

ADK Sequential agents: the **Analyst** gathers facts with real tools; the **Advisor** turns them into guidance.

### Answers the brief end to end

A real ingest, clean, model, visualize pipeline; 2+ Google Cloud items; 2+ NVIDIA items; a genuinely useful output.

### Built to scale beyond the hackathon

The same architecture serves any city or pollutant dataset; just repoint the ingestion.

## List of features offered by the solution

### Live multi-city dashboard

Delhi, Mumbai, Kolkata, Chennai: 90-day trends, 7-day rolling averages

### Safety Score (0 to 100)

one plain-language verdict per city, driven by WHO exceedance

### Human Impact panel

cigarette-equivalent exposure and life-expectancy impact

### Pollution hotspot ranking

the worst stations in each city, ranked by average PM2.5

### Ask VayuSense chat

ADK multi-agent pipeline (Analyst → Advisor) on Gemini

### GPU-accelerated analytics

NVIDIA RAPIDS cuDF backend with a live 37.5× benchmark panel

### WHO-guideline comparisons

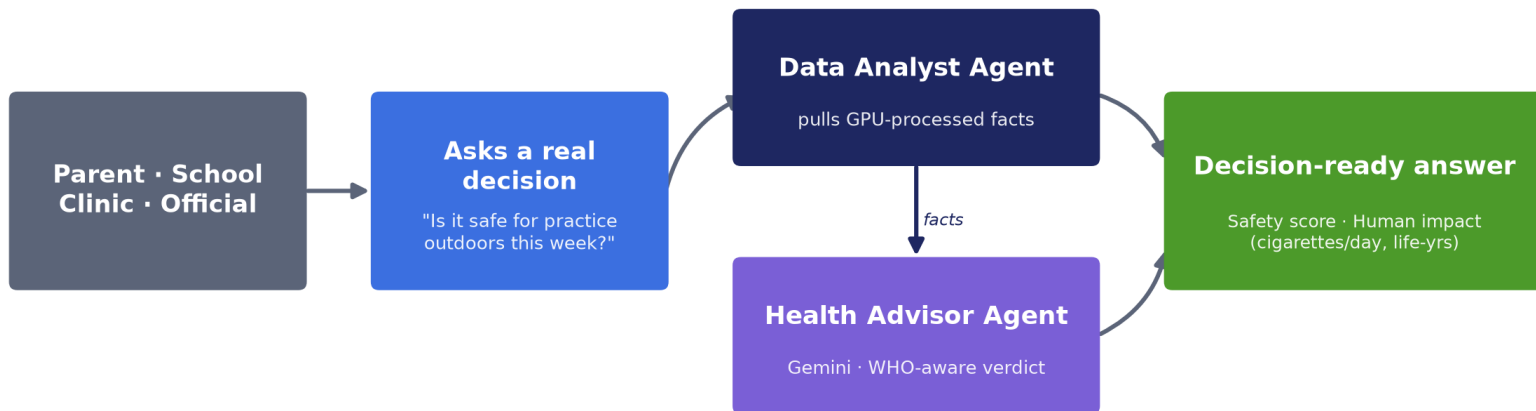
six pollutants tracked against WHO 24-hour limits, with trends

### Deployed and public

live right now at [vayusense-kggf.onrender.com](https://vayusense-kggf.onrender.com)

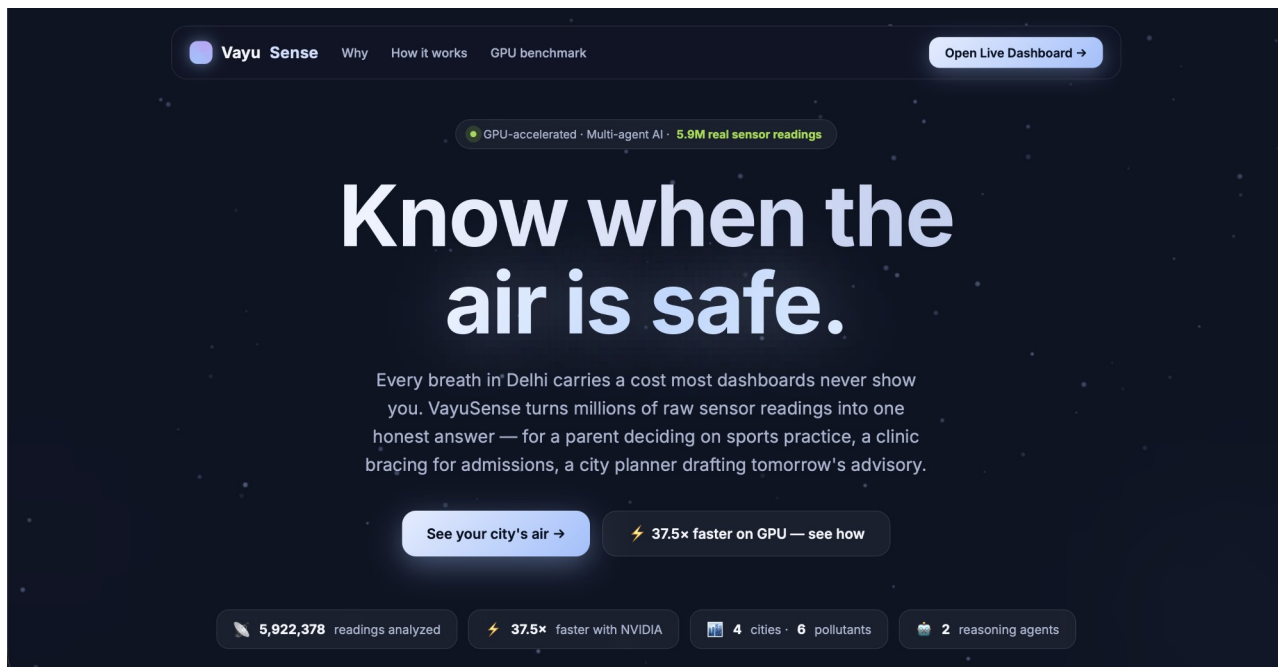
## Process flow diagram or Use-case diagram

### VayuSense: Sequential Multi-Agent Pipeline (Google ADK + Gemini)



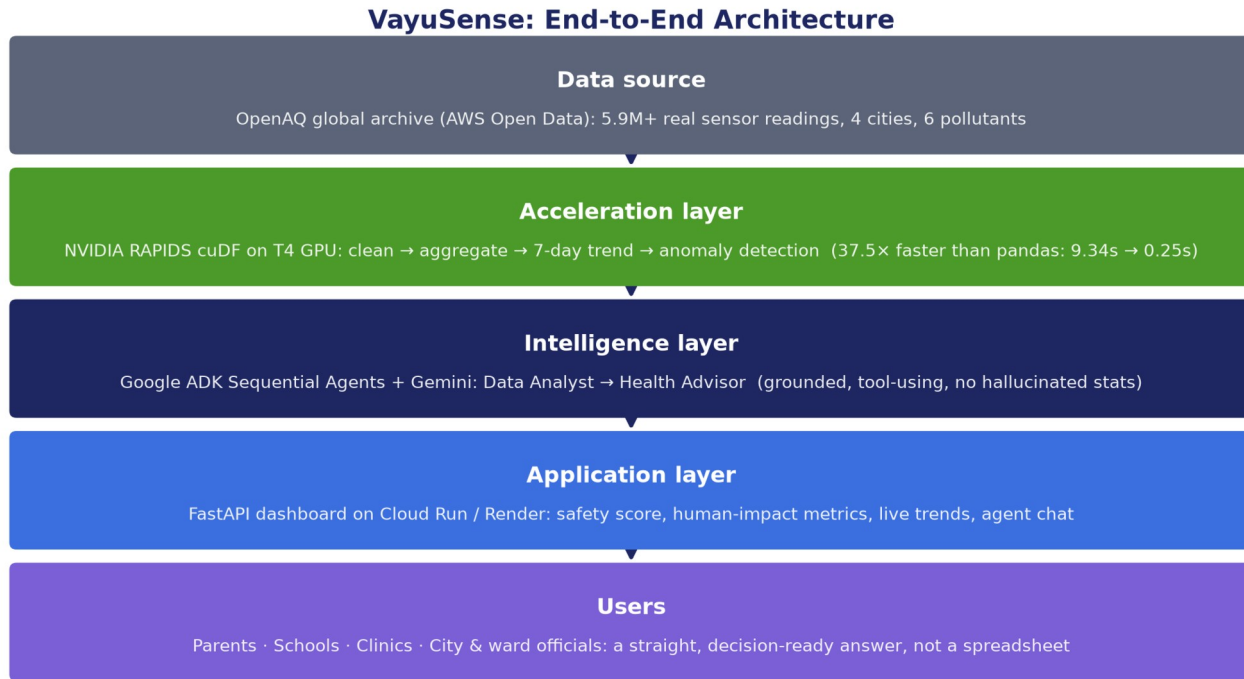
One question in, one decision out: the ADK pipeline grounds every answer in GPU-processed data.

## Wireframes/Mock diagrams of the proposed solution



Landing page of the deployed product: [vayusense-kggf.onrender.com](https://vayusense-kggf.onrender.com)

## Architecture diagram of the proposed solution:





## Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

### Google Cloud

**Gemini 2.5 Flash** powering both agents' reasoning

**Google ADK** Sequential multi-agent orchestration, tool calling

**Cloud Run-ready** containerized FastAPI app (Docker)

### NVIDIA

**RAPIDS cuDF** GPU dataframes: clean, aggregate, trend, anomaly

**T4 GPU** (Google Colab) as the benchmark target

**9.34s → 0.25s** on 5.9M rows: a 37.5× measured speedup

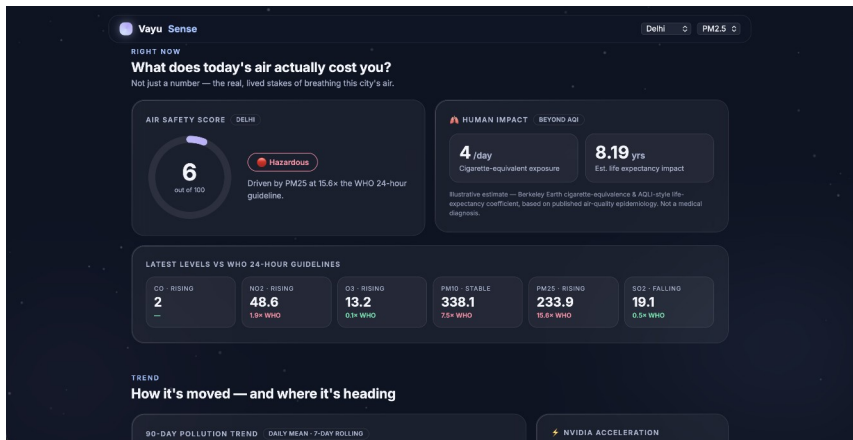
### Why this stack

**Speed:** millions of sensor rows need cuDF-class processing to stay decision-fresh instead of report-stale.

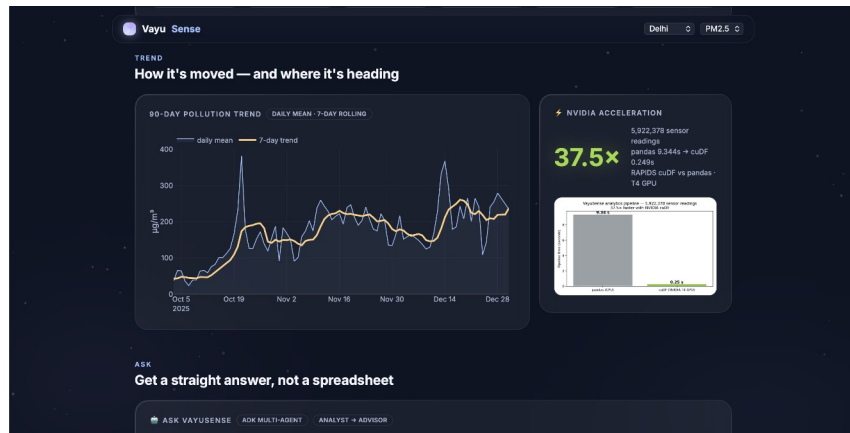
**Trust:** grounded, tool-calling ADK agents avoid the hallucinated health claims a single freeform prompt risks.

**Portability:** one lightweight container deploys to Cloud Run, Render, or on-prem with no code changes.

## Snapshots of the prototype



Safety Score, WHO comparisons and Human Impact, live for Delhi



90-day trends, the live GPU benchmark and the agent chat

Try it live: [vayusense-kggf.onrender.com](https://vayusense-kggf.onrender.com)



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# Thank you

